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Neogene deformation of the flat-subduction segment of the Argentine-Chilean Andes: Magnetostratigraphic constraints from Las Juntas, La Rioja province, Argentina

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ABSTRACT

Substantial Neogene deformation occurred in the part of the Andes Mountains and the Sierras Pampeanas that overlie a subhorizontal segment of the subducting Nazca plate in northern Argentina. The history of the volcanic arc suggests that the Nazca plate attained the present low-angle inclination between about 17 and 10 Ma, with the region underlain by the flat Nazca plate widening eastward thereafter. We compare the history of deformation, provided by magnetostratigraphy of Miocene strata in foreland and intermontane basins, to the history of the subducting plate.

At Las Juntas, in the Precordillera thin-skinned thrust belt, 4,780 m of Neogene nonmarine strata record the initiation and termination of sedimentation in a segment of a foreland basin. Magnetostratigraphic data, complemented by fission-track dating of an interbedded tuff, indicate that the strata span from 18 to ~9 Ma. The earliest strata correlate to the end of a time of volcanic activity in the magmatic arc and the beginning of reverse faulting in the Frontal Cordillera. Subsequently, the deformation front migrated eastward, generating the Precordillera thrust belt, and finally the Sierras Pampeanas basement uplifts. The rate of strata accumulation varied systematically as new uplifts arose, with a maximum rate exceeding 1.0 mm/a between ~15 and 12 Ma.

A correlation between the deformational history and the geometric history of the subducted slab suggests that thrusting migrated eastward as the angle of subduction shallowed. Although large-scale factors affecting the plate margin beyond the flat-slab segment may have partly induced the compressional deformation, this correlation suggests that processes related to flat subduction are of importance in deforming the continental plate.

INTRODUCTION

A causal relationship may exist between the down-dip geometry of a subducted plate and the style and distribution of deformation in the lithospheric plate above it (for example, Cross and Pilger, 1982). The Andes Mountains system is an excellent region in which to study this possibility, because the subducting Nazca plate is divided into several major segments of contrasting dip (Barazangi and Isacks, 1976; Isacks and Barazangi, 1977; Jordan and others, 1983a). The focus of this paper is the flat-subduction segment of central Chile and Argentina between 28° and 33°S where a 600-km-wide zone of the Andes Mountains and foreland has been shortened during the Miocene to Recent (Jordan and Allmendinger, 1986) (Fig. 1). A similar situation, of flat subduction and spatially coincident shortening, existed in North America during the Late Cretaceous and early Cenozoic (Cross and Pilger, 1978; Dickinson and Snyder, 1978; Engebretson and others, 1984), for which Bird (1984, 1988) recently proposed a mechanical model that links the deformation in the Rocky Mountains to shear above a shallowly subducting plate. The main objective of this paper is to present new chronological data that constrain the sequence and timing of deformation of the Chilean and Argentine Andes Mountains in the zone of flat subduction. The product is a comparison of the history of deformation to the history of the subducting plate, without presupposing cause and effect relationships, but the focus is on the task of documenting the progression of deformation through space and time. The deformation history is derived from the history of strata that accumulated at Las Juntas (La Rioja province), part of a foreland basin on the eastern flank of the mountain belt.

This is the second region in the Bermejo foreland basin in which the chronology, provenance, and depositional history of the basin-filling strata have been documented (Johnson and others, 1986; Beer and Jordan, 1989; Damanti, 1989). A significant difference between the new Las Juntas site and the previous Huaco site is that Las Juntas is located several thrust sheets west of Huaco, thus representing the history of a more interior part of the orogenic system.

Segmentation of the subducted Nazca plate, which is clearly expressed in earthquake patterns, apparently either formed or became more accentuated during the middle Miocene (Jordan and others, 1983b). This conclusion is based largely on the observed relationship between activity of a magmatic arc and the subduction angle. The flat-slab segment of Chile-Argentina (28°S–33°S) is characterized by its lack of an active volcanic arc, whereas there is an active magmatic arc in other segments beneath which the subducted plate is more steeply inclined (Barazangi and Isacks, 1976; Jordan and others, 1983a). The correspondence between geometry of the modern subducted plate and presence or absence of an active volcanic arc is the basis for the working hypothesis that variations in magmatic activity through time are indicative of the paleogeometry of the ancient subducted plate.

On the basis of this premise, the history of segmentation and dip angle of the Nazca plate can be derived from the chronology of magmatic activity. A magmatic arc existed in the modern flat-slab segment during the early and

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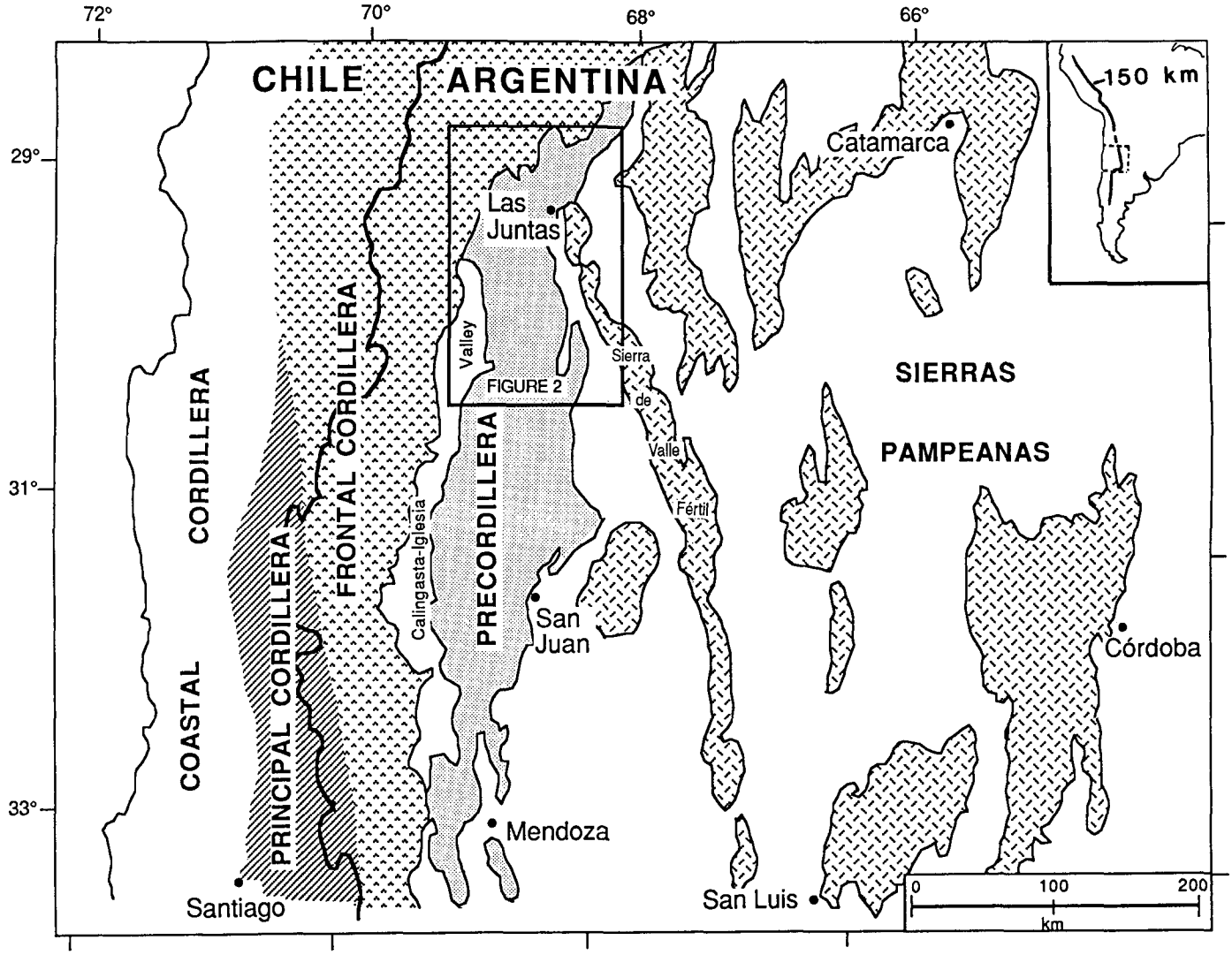


Figure 1. Map of the flat-slab segment of Chile and Argentina. The box indicates the location of Figure 2. Inset map (upper right) shows position of regional map relative to South America and the 150 km contour of the Wadati-Benioff zone (map by R. Allmendinger after Barazangi and Isacks, 1976).

middle Miocene, whereas only minor magmatism has occurred since about 10 Ma. Much of the young magmatism has occurred several hundred kilometers to the east of the older volcanic line (for example, Munizaga and Vicente, 1982; Maksaev and others, 1984; Kay and others, 1988; Tabbutt and others, 1990). This implies that the subducted plate was more steeply inclined before 10 Ma than it is now (Jordan and others, 1983b). Furthermore, geochemical evidence suggests that the subducted-plate angle began to decline after about 17 Ma, long before the magmatism ceased (Kay and others, 1987; Ramos and others, 1990).

At the surface, the Chile-Argentina flat-slab segment is characterized by two styles of Neogene deformation: thin-skinned thrusting on the eastern flank of the high Andes, and basement thrusting in a broad foreland to the east (Fig. 1) (Jordan and others, 1983a; Jordan and Allmendinger, 1986; Jordan and Gardeweg, 1989). Both styles of compressional deformation were coincident in a broad sense with the flattening of the descending plate. In addition, the partial coincidence of basement uplifts in the foreland with the limits of flat subduction suggests that basement deformation is somehow driven by flat subduction (Cross and Pilger, 1982; Jordan and others, 1983a; Bird, 1984, 1988; Jordan and Allmendinger, 1986).

The structural provinces of the flat-slab region are distinct from those of segments to the north and south, but Miocene to Recent compressional deformation is not limited to this region. Thus, at least some processes that link subduction to deformation may operate at a scale that exceeds that of the individual segments (Isacks, 1988; Jordan and Gardeweg, 1989). The first step toward understanding the cause and effect relationships between plate geometry, magmatism, and deformation is to improve the resolution of the spatial and temporal trends of deformation.

A reliable, precise chronology of synorogenic Cenozoic strata can be used to unravel the complex structural relationships that evolved as a result of Nazca plate segmentation. Cenozoic strata were sequentially produced and subsequently deformed by the deformation and uplift of the Andean ranges. Because the Cenozoic strata are widely distributed, crosscutting rela-

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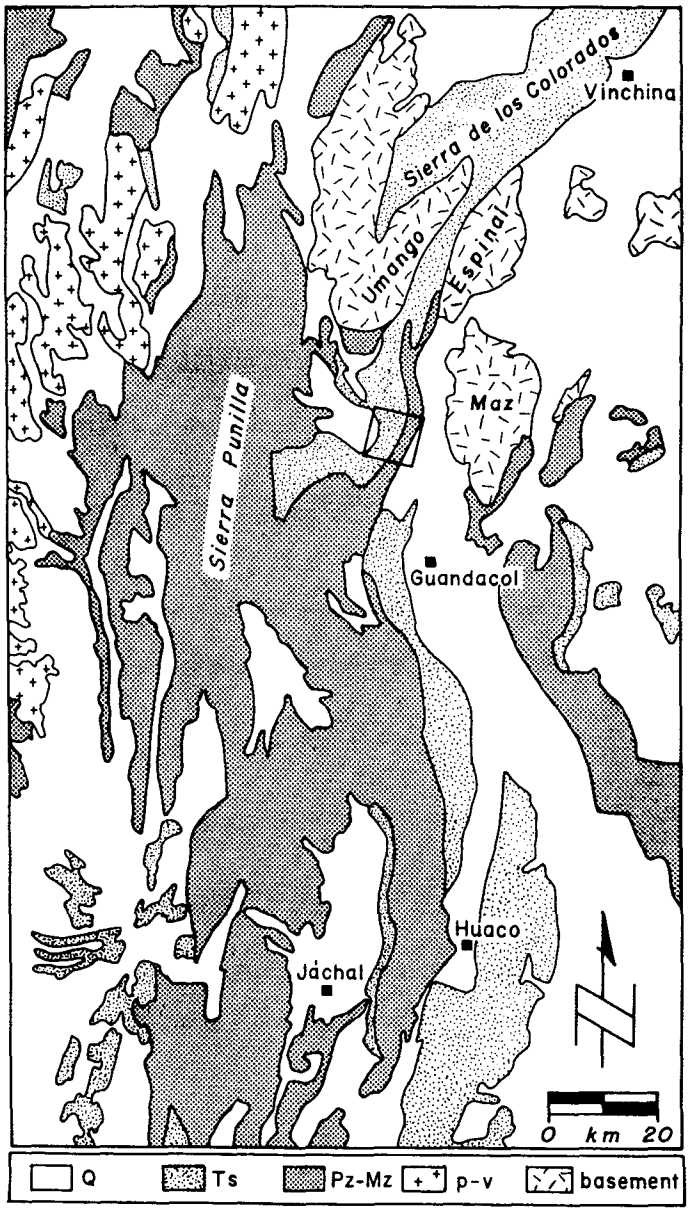


Figure 2. Bedrock geology of the structural provinces bordering the Las Juntas site. See Figure 1 for location. Q, Quaternary cover; Ts, Tertiary strata; Pz-Mz, sedimentary units of Paleozoic and Mesozoic age; p-v, plutonic and volcanic rocks of Paleozoic and Mesozoic age. Compiled from Furque (1972a, 1979), Turner (1964), Gentili (1972), and Fielding and Jordan (1988). The box indicates the location of Figure 3.

tions between those strata and faults or folds have broad application to dating of the deformation. In addition, studies of the rate and timing of the accumulation of the foreland basin strata offer a less direct means to date deformation. The interpretations that follow are based on premises concerning the relationship between (1) thrust shortening (causing tectonic loads) and subsidence, (2) clast petrology and source-area uplift, and (3) facies character and proximity to uplifts (for example, Steidtmann and Schmitt, 1988; Jordan and others, 1988).

The focus of this investigation is a nearly 5-km-thick sequence of Neogene continental strata at Las Juntas in the western part of La Rioja province, Argentina (Figs. 2 and 3). We examine the uplift of the main Andean range (Frontal Cordillera at this latitude) and the thin-skinned Precordillera thrust belt. Magnetic polarity stratigraphy of continental sediments shed from the rising mountain ranges and a fission-track date from an interbedded volcanic layer are used for geochronometry. Crosscutting relations between the Las Juntas section and structures indicate times of local deformation. Variations through time in the petrology of gravel and sand grains indicate times of deformation west of Las Juntas. The Las Juntas section is compared to other Neogene sections in the region for which geochronological, sedimentological, and petrological data are also available. From these comparisons, which include ages of key horizons, rates of stratal accumulation, and sedimentological information, further insight is gained into the timing of deformation in regions adjacent to Las Juntas. More detailed information about the field and laboratory procedures may be found in Reynolds (1987). More detailed information about sandstone petrology may be found in Damanti (1989).

STRUCTURAL SETTING

Three structural provinces influence the strata exposed at Las Juntas (Fig. 1): the major Andean range (Frontal Cordillera), a thin-skinned fold-thrust belt east of the Frontal Cordillera (Precordillera), and an eastern province of thick-skinned basement uplifts (Sierras Pampeanas). The bedrock compositions of the structural provinces are sufficiently different to allow identification of the general source areas of sand and gravel clasts.

Frontal Cordillera

The Frontal Cordillera, along the Chile-Argentina border, is the drainage divide between the foreland and forearc systems. It is composed of thick accumulations of extensive Carboniferous to Triassic volcanic and plutonic rocks (Caminos, 1972). Carboniferous clastic strata are important locally, and early and middle Miocene volcanic units are widespread (Maksaev and others, 1984).

Reverse faults in the Frontal Cordillera with important vertical throw formed during the Miocene, but there is little evidence of the large-scale horizontal shortening (Maksaev and others, 1984; Moscoso and Mpodozis, 1989; Jordan and Gardeweg, 1989). Recent mapping in Chile determined that most of the faulting occurred between 18.9 and 16.6 Ma (Maksaev and others, 1984), whereas field relations in Argentina imply that faulting occurred between ~11 and 6 Ma (Ramos and others, 1987, 1990). Younger deformation appears to have been confined to regional uplift, without causing significant differential movement on faults that break the surface.

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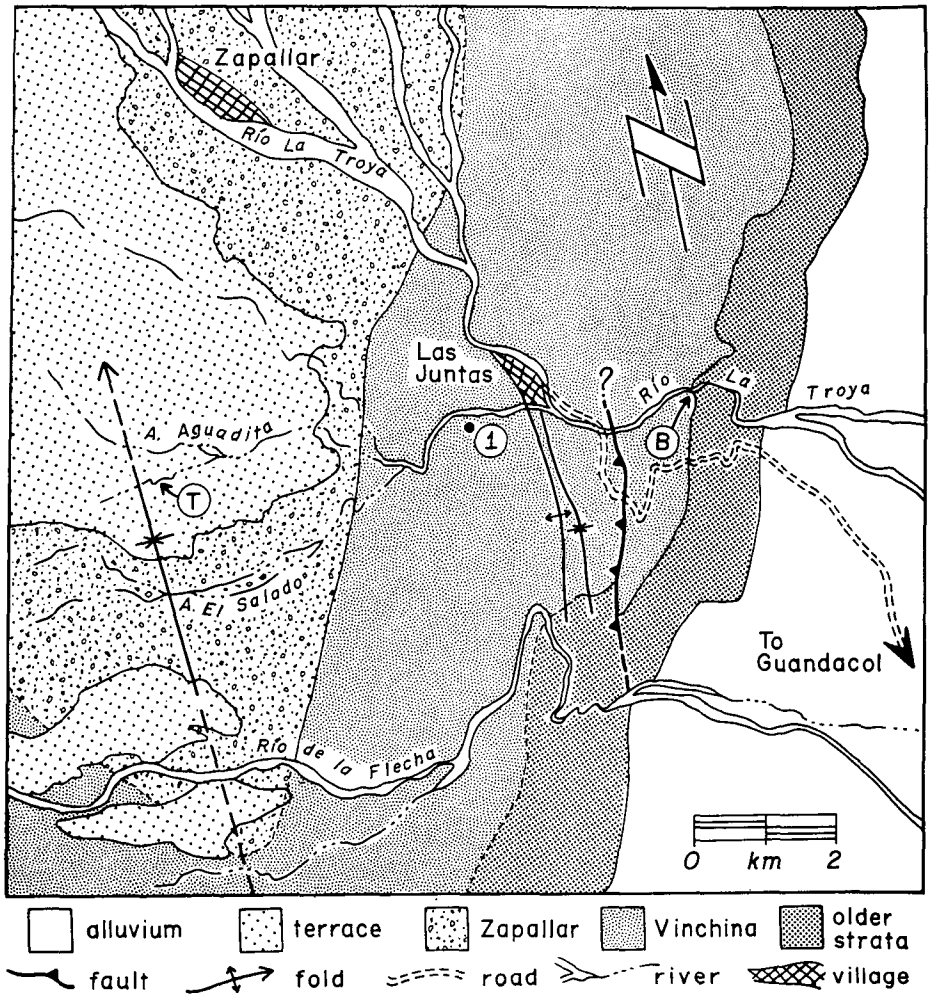


Figure 3. Geologic map of the Las Juntas area, showing base (B) and top (T) of the section measured and sampled for magnetic polarity stratigraphy. The circled 1 is the location of the dated tuff interbed. Location indicated in Figure 2.

Precordillera Fold-Thrust Belt

The Precordillera is a 70- to 120-km-wide, predominantly east-verging fold-thrust belt separated from the Frontal Cordillera by the longitudinal Calingasta-Iglesia Valley (Baldis and Chebli, 1969; Furque, 1972b; Ortiz and Zambrano, 1981). The Precordillera exists only above the southern and central parts of the flat-slab subduction segment; the characteristic thrust geometry pinches out to the north of the Las Juntas study area (Fig. 1).

The Precordillera exposes Paleozoic marine strata, local basalts and andesites, and upper Paleozoic and Triassic nonmarine clastic strata. South of the study site, there is a distinctive carbonate unit at the base of the thrust sheets of the central and eastern part of the Precordillera (for example, Baldis and Chebli, 1969), but at the latitude of Las Juntas, only clastic strata are exposed (Furque, 1972b). Deformed Neogene nonmarine strata are the most common unit exposed in the eastern part of the Precordillera, but they are considerably less common in the central and western parts of the Precordillera.

Sierras Pampeanas

Thick-skinned, crystalline basement uplifts of the Sierras Pampeanas occur across a 350- to 400-km-wide area between 26° and 33°S. These ranges were uplifted by rotation and shortening above moderately dipping thrusts during the late Miocene, Pliocene, and Quaternary (González Bonorino, 1950; Jordan and others, 1983a, 1983b; Jordan and Allmendinger, 1986; Reynolds, 1987; Tabbutt and others, 1990).

The Sierra de Maz, a northern extension of the Sierra de Valle Fértil range, is located less than 10 km east of the Cenozoic exposures of Las Juntas (Fig. 2). The Sierra de Valle Fértil system is soled by a west-verging thrust (Ortiz and Zambrano, 1981; Snyder and others, 1986). Two other Pampean blocks, the Sierra de Espinal and Sierra de Umango, constrict the northward continuity of the Las Juntas section and deform the strata described here (Fig. 2) (Furque, 1972a).

The ranges of the northwestern Sierras Pampeanas are composed predominantly of Precambrian metamorphic and plutonic rocks. In the area bordering the Las Juntas region, thick Permian-Triassic red beds covered the basement before deformation began in the Miocene (Stipanicic and Bonoparte, 1972).

The boundary between the Precordillera and Sierras Pampeanas structural provinces is, in most places, buried beneath the Quaternary fill of the Bermejo Valley (for example, Fielding and Jordan, 1988). The northern trend of the Precordillera and the northwestern trend of the western Sierras Pampeanas, however, intersect at about 29°S (Furque, 1972a, 1972b), immediately north of the study site at Las Juntas (Fig. 2). The nature and timing of interactions between the two structural domains in the area of intersection are not yet well studied.

STUDY SITE

Our work was carried out near Las Juntas, located in the Cuenca de La Troya, in the northern part of the central Precordillera (Figs. 2 and 3). A dirt road from the village of Guandacol provides access to the section.

Furque (1972a) mapped the area at a scale of 1:200,000 and described the Tertiary stratigraphy. The strata are exposed in a north-trending, north-plunging syncline. Our magnetostratigraphic section is located in vertically continuous, west-dipping strata incised by the Río de La Troya and lower-order tributaries on the east limb of the syncline (Fig. 3). The strata that we studied unconformably overlie eolian strata of the El Aspero Formation, which was assigned a Triassic(?) age by Furque (1972a). The rocks at Las Juntas were deposited in a subsiding foreland basin, unlike other exposures of Tertiary strata to the west and south (Fig. 2), which accumulated in piggyback basins formed during uplift of the Precordillera (J. A. Beer, unpub. data).

METHOD OF INVESTIGATION

Chronological determinations are based on magnetostratigraphy data, determined from a total of 142 stratigraphic levels in a 4,780-m sequence of detrital strata, and a fission-track date. The provenance history is interpreted from 38 samples of sandstone for which modal analyses were performed, combined with qualitative field estimations of compositions of conglomerates encountered in numerous horizons. Facies determinations are based upon less detailed information, particularly architectural styles, tex-

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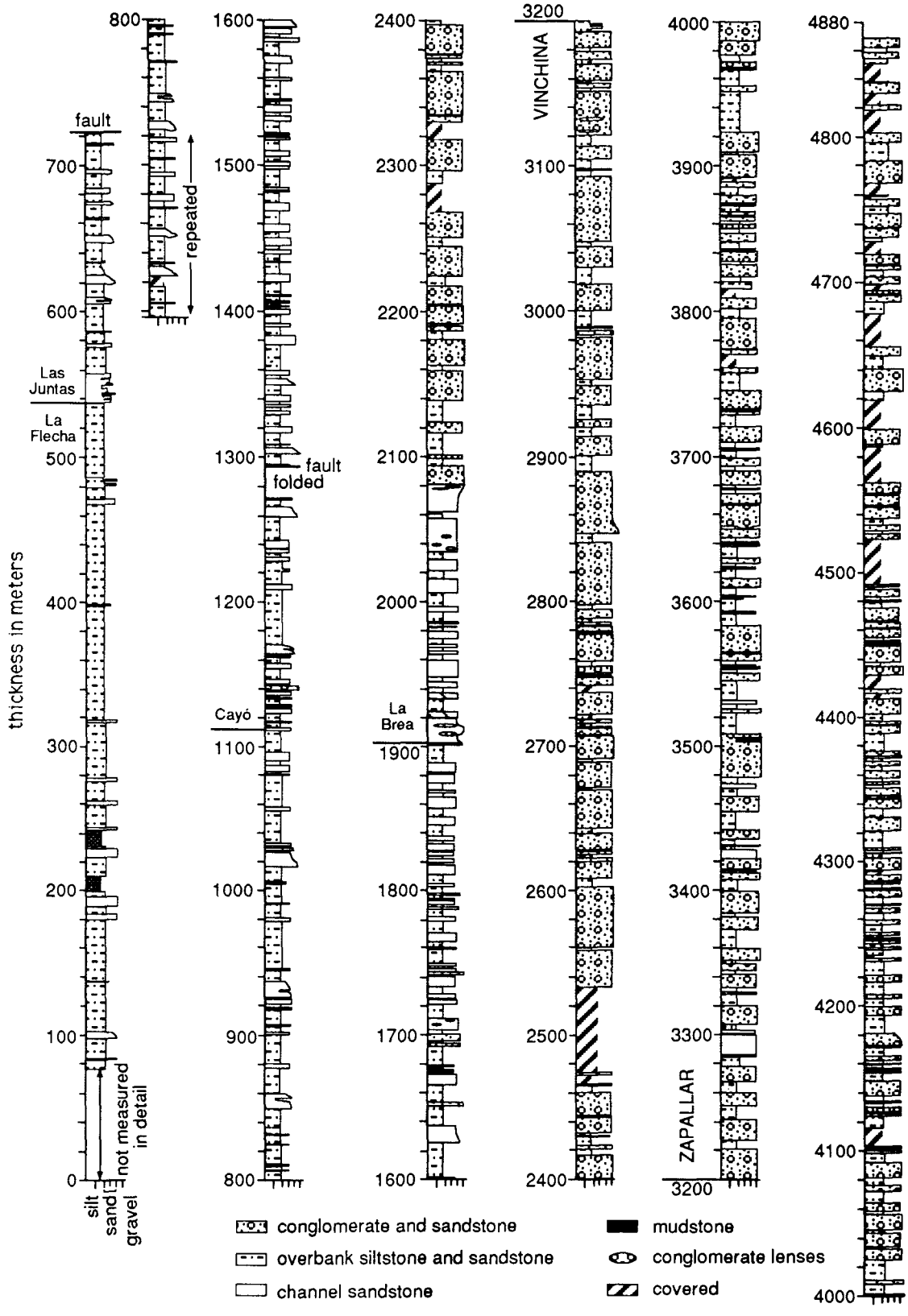


Figure 4. Lithologies at Las Juntas measured along Río La Troya and its tributaries (see Fig. 3). A simplified version of this information constitutes the columns in Figure 5, on which site levels are indicated. Informal member designations (J. F. Damanti, unpub. data) are indicated.

tures, and sedimentary structures, collected while measuring and describing the entire section (Fig. 4). The section studied was selected for vertical continuity of exposure, a prerequisite for successful magnetostratigraphy, rather than for lateral continuity or three-dimensional exposures, a convenience for facies studies.

Sample collection for purposes of magnetic polarity stratigraphy followed the procedures established by Johnson and others (1975). A minimum of 3 fist-sized, oriented samples was taken at each of 168 sites. At seven of these sites, a single, larger block sample was collected and cut into smaller pieces in the laboratory. Sites were sampled at approximately 30-m intervals. Only

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siltstones and mudstones were collected (Fig. 5) in the hope that the magnetic minerals would be of sufficiently fine-grained size to deliver a reliable magnetic signal. The 4,780-m section was described and measured using a tape measure and Brunton compass while the initial sampling was in progress.

Samples were milled into 1-in. "cubes" on a grinding wheel with carborundum grinding disks. Magnetic orientations were determined using a Schonstedt SSM-1 fluxgate spinner magnetometer.

The magnetic properties and mineralogy of four sites were inferred using standard systematic stepwise thermal and alternating-field demagnetization methods (Fig. 6) (Collinson, 1983). Comparison of the temperature or alternating field strength to the percentage of the maximum magnetic moment indicates that magnetite is the primary magnetic mineral in all of the lithologies studied and that hematite is also a magnetic constituent in three of the four sites (Reynolds, 1987). Orthogonal progressive demagnetization diagrams (Zijderveld, 1967) show, in general, that the rocks have a single component of magnetization with a very weak northward Brunhes viscous overprint magnetization that was easily removed in the first demagnetization steps (Fig. 6). The primary component of magnetization is interpreted as a detrital remanent magnetization (DRM).

All samples in the section were first analyzed to determine their outcrop natural remanent magnetization (NRM). The NRM magnetic dipole moment/unit volume (M in amperes per meter) for the Las Juntas strata varied between 2.1×10^{-3} and 6.0×10^{-1} A/m, with most falling in the 7.7×10^{-3} to 6.5×10^{-2} A/m range (1 A/m corresponds to a cgs magnetization of 10^{-3} emu/cm^3; Shive, 1986).

Because magnetite is the principal carrier of depositional remanence, an initial thermal treatment of 400–500 °C was often sufficient to isolate the primary remanent component; however, about 40% of the sites yielded ambiguous mixed polarities or had suspicious low-inclination normal polarities after the initial thermal treatment. These received additional thermal demagnetization until the primary component was isolated and Class I results were achieved. As used here, a Class I site refers to results that satisfy Watson's criteria (R) for nonrandomness at the 95% confidence level. A minimum of three samples per site (N) is required to achieve

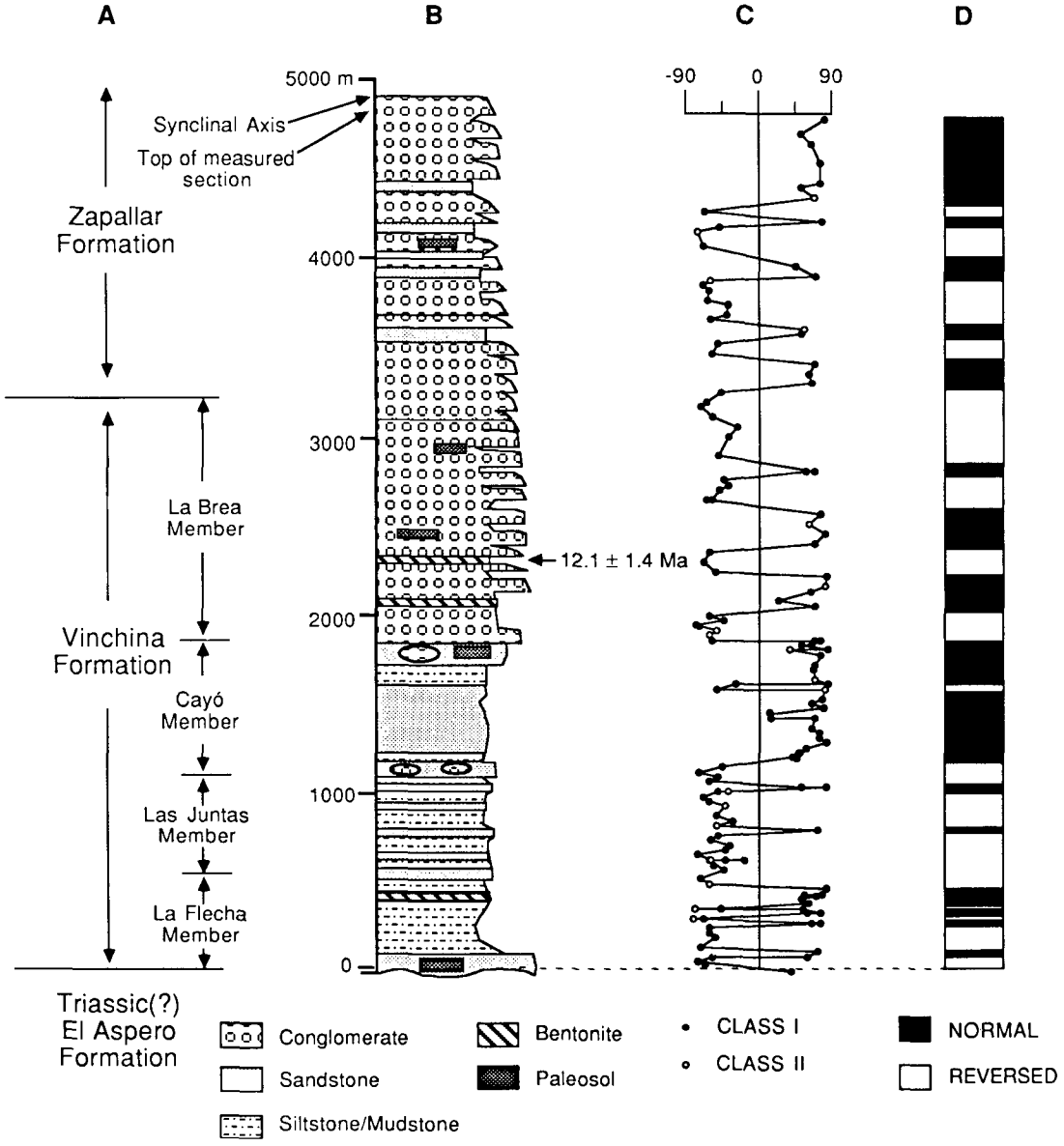


Figure 5. Stratigraphic, facies, and magnetostratigraphic summary of the Las Juntas section. Column A shows the formation boundaries defined by Furque (1972a) and the informal members used in this study. Column B outlines the lithostratigraphy (simplified from Fig. 4). Column C is a plot of VGP latitude determined from each sample site versus stratigraphic level of the site; the VGP data are abstracted to a local magnetostratigraphic column (column D).

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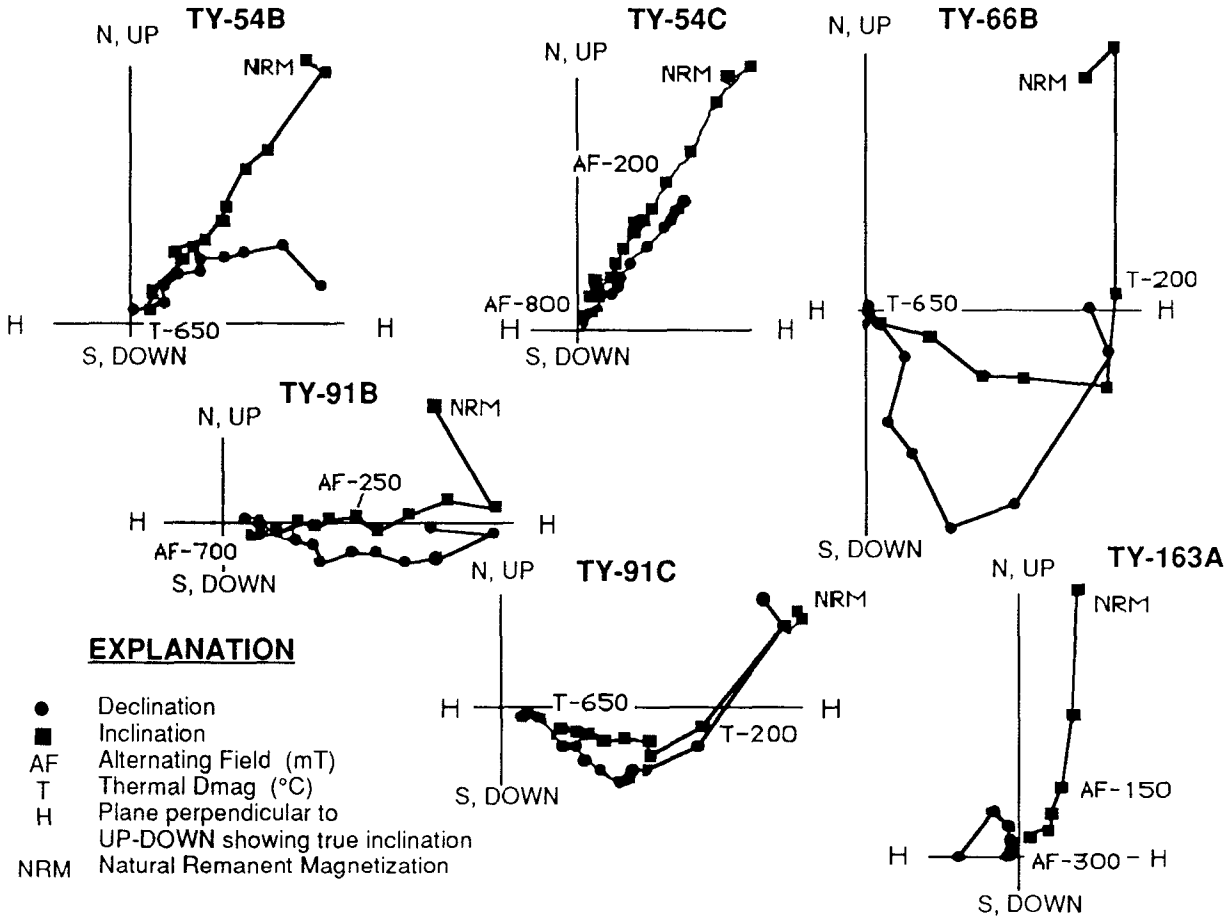


Figure 6. Zijderveld (1967) plots of selected samples treated by stepwise demagnetization. A weak, northward Brunhes viscous overprint was removed in the early stages of the demagnetization procedure. Samples TY-54B and TY-54C (1,490 m level) are from normal Anomaly 5AD; TY-66B (1,900 m level) is from the reversed interval between Anomalies 5AB and 5AC; TY-91B and TY-91C (3,040 m level) are from the reversely polarized interval between Anomalies 5A and 5AA; TY-163A (3,890 m level) is from a reversed interval between Anomalies 5 and 5A. AF demagnetization steps are expressed in SI units of induction ($\bar{B}$ in teslas). An induction of 1 T corresponds to a cgs magnetic field $\bar{H}$ of 10^4 Oe (Shive, 1986).

Class I results. For N = 3, it is required that R > 2.62 (Watson, 1956). Class II data are those for which (1) the results may not be statistically significant but the polarity of all samples from the site is the same or (2) only two samples survived transport from field to laboratory. Class II data are included in the stratigraphic column only to support adjacent Class I sites. Sites from which the data did not conform to these criteria were rejected from the sample set.

Fission-track ages were determined using the external detector method on zircons collected from volcanic tuffs interbedded with the basin strata. The zircons were separated using standard heavy-liquid and magnetic separation techniques. Sample preparation and counting procedures used are described by Naeser (1978).

Modal compositions of fine- to medium-grained sandstones were determined using a petrographic microscope from thin sections stained for calcium and potassium (Houghton, 1980). The Gazzi-Dickinson method of point-counting was used (Ingersoll and others, 1984). The composition of conglomerates was recorded as relative abundance of major clast compositions.

STRATIGRAPHY

Furque (1972a) divided the strata that we studied into two formations; he lacked fossil control but assigned the formations to the Tertiary. Our new data constrain the ages of the Vinchina and Zapallar Formations at Las Juntas to the Miocene. We attempt to utilize Furque's stratigraphic subdivisions but note that the contact between the two formations coincides with a general change in the exposure characteristics, rather than with an easily recognized change of depositional facies. The Vinchina Formation is more resistant than the Zapallar Formation, and the contact is gradational.

Vinchina Formation

Furque (1972a) assigned the lower unit at Las Juntas to the Vinchina Formation, which Turner (1964) and Ramos (1970) described at its type locality ~75 km to the northeast in the Sierra de los Colorados (Fig. 2). Several vertebrate fossil discoveries (Ramos, 1970) substantiated Turner's Miocene age assignment in the Sierra de los Colorados. Two fission-track dates in the Vinchina Formation (Table 1 and in Tabbutt, 1987; Tabbutt and others, 1987a) from the Sierra de los Colorados establish that the upper 3,200 m were deposited between 7.3 ± 1.2 and 4.3 ± 1.0 Ma (errors at 2σ).

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TABLE 1. FISSION-TRACK DATA

Sample	Number of grains	Fossil track density (tracks/cm^2)	Induced track density	Age (Ma)	Error (2σ)
TY2 (Las Juntas)	5	1.8×10^6	8.3×10^6	12.1	2.8
VN2* (Vinchina)	6	6.1×10^5	7.8×10^6	4.3	1.1
VN3* (Vinchina)	3	5.2×10^6	4.3×10^7	7.3	1.4

Constants: $\lambda f = 7.03e-17$, $\lambda d = 1.551e-10$, $\sigma = 5.80e-24$, $I = 7.252e-3$.
*Tabbutt (1987) and Tabbutt and others (1987a).

At Las Juntas, the Vinchina Formation is about 3,200 m thick and gradually coarsens upward (Fig. 4). In the Sierra de los Colorados, the average thickness is nearly 4,450 m, and it is subdivided into two mappable members (Ramos, 1970). We denote major depositional environment units in the Vinchina Formation at Las Juntas with four informal members.

La Flecha Member. The basal 60 m consists of coarse sandstone with meter-scale bedforms; a rotated block of laminated fine sandstone embedded in coarse sandstone implies that erosional relief at (unexposed) channel margins was as much as 2 m. The basal coarse beds grade to well-indurated reddish-brown siltstone with interbedded fine-grained sandstone, which are the predominant rock types upward to approximately 540 m (Figs. 4 and 7A). Evidence of soil formation, such as calcite-filled vugs, is common below ~250 m.

Las Juntas Member. Above 540 m, medium- to coarse-grained, thickly bedded sandstone bodies (Fig. 7B) become common. The sandstone bodies are composites of several sandstone beds with thin siltstone interbeds, typically fining upward and with bed thicknesses diminishing upward. Locally the sandstone beds ($\leq$15 cm thick) define lateral accretion surfaces. The sandstone bodies are characterized by erosional bases, intraclasts, horizontal lamination, and cross-bedding. Lateral discontinuity is one of the first-order characteristics of the unit. It occurs on a scale of 5 to 10 m near the middle of the Las Juntas member to as much as 50 to 100 m at the top of the member. Relief on the bases of sandstone bodies is as much as 2 m. The sandstone bodies are interbedded with siltstone and thin, lensing intercalated sandstone (Fig. 7C). These have small-scale cross-bedding and ripples, mudcracks, and common trace fossils. Sandstone bodies gradually increase in thickness upward (Fig. 4).

Cayó Member. Above 1,100 m, the strata are similar to those of the Las Juntas member except that the sandstone bodies are locally conglomeratic, and they are clustered into thicker series (Fig. 7D).

La Brea Member. By about 1,900 m, sandstone bodies are multi-story sheets (Fig. 7E); pebble and cobble conglomerate lenses and beds are abundant. Interbedded siltstone and fine sandstone beds commonly contain evidence of soil development.

The gradational contact between the La Brea member of the Vinchina Formation and the Zapallar Formation is placed at 3,200 m. Conglomerate beds are minor components even at the base of the section but are abundant above the 2,100 m level. The scale of all of the facets of the rocks (grain size, bed thickness, bedforms, lateral continuity of beds) increases upward.

Approximately 1,000 m of strata exposed 15 km north of the measured section, on the southeast flank of Sierra Umango (Fig. 2), is thought to correlate to the Las Juntas and Cayó members. This association is based upon the correlation of an undated thick bentonite near the base of the northern section to a thick bentonite at the 400 m level (upper La Flecha member) in the measured section. By ~500 m above the bentonite, cobble-sized clasts occur sporadically in sandstones, and laterally continuous conglomerates occur above a level ~600 m above the bentonite. The bentonite at 400 m in the measured section could not be physically traced to the conglomeratic section owing to long distances, rugged relief, and deformation, but it is the only recognized bentonite in the lower 1,500 m of the measured section (Fig. 5) and is persistent over several kilometers north and south of the measured section. In addition, the possibility of significant northward coarsening is consistent with slight northward coarsening over a distance of about 2 km from the measured section.

Zapallar Formation

The Zapallar Formation, named by Furque (1972a), conformably overlies the Vinchina Formation. Furque (1972a), however, did not correlate the Zapallar Formation with the Toro Negro Formation, the unit that overlies the Vinchina Formation in the Sierra de los Colorados (Turner, 1964).

In the Las Juntas section, the Zapallar Formation is nearly 1,600 m thick, with its upper contact eroded (Fig. 5). It is dominated by repeated fining-upward bodies of conglomerate and medium- to coarse-grained sandstone, with sheet geometries; these pinch out laterally over distances ranging from tens to hundreds of meters. Subangular to rounded clasts are pebble to cobble sized, interbedded at 10-cm to 2-m scale with sand layers; horizontal lamination dominates over large-scale cross-stratification, but both are common. Conglomerate beds occur as sheets (Fig. 7F) and lenses within the larger-scale conglomerate bodies; bases of the conglomerate bodies have at least 2 m of erosional relief. Large intraclasts of siltstone and sandstone, typical of the interbedded finer-grained deposits, are common. Imbricated clasts occur locally. Interbeds of moderately to poorly sorted sandstone are thicker and have greater lateral continuity (Fig. 7G) than do the analogous sandstone and siltstone deposits of the upper part of the Vinchina Formation. The poorly sorted sandstones typically lack primary depositional structures and locally have caliche nodules, suggesting soil development.

A 30-m interval for paleomagnetic samples could not be maintained in the upper part of this unit because poorly sorted sandstone replaces siltstone in the finer-grained intervals and because finer-grained deposits are poorly exposed near the top of the section. Exposure deteriorates westward beneath Quaternary terraces, such that, from the top of the measured section to the axis of the syncline, only discontinuous outcrops of conglomerates and sandstones occur (Fig. 3).

Provenance

Conglomerate clasts in the Zapallar and Vinchina Formations are similar. Beginning at 1,140 m, pebbles consist predominantly of green sandstone clasts, but clasts of granitoids and intermediate to siliceous volcanic rocks are also common. These rock types are known in Paleozoic and Mesozoic units of the Precordillera west of Las Juntas and in the Frontal Cordillera (Fig. 2) (Furque, 1972a; Ortiz and Zambrano, 1981; Reynolds and others, 1986, 1987a). Metamorphic clasts (gneisses, biotite schists, muscovite schists, and garnet schists) occur in strata 15 km north of the measured section that are thought to correlate with the Las Juntas and Cayó members; the metamorphic compositions are typical of the crystalline basement exposed in the immediately adjacent Sierras de Maz, Espinal, and Umango (Fig. 2), but imbricate clasts indicate paleoflow from the west to east.

Modal analyses of 38 fine- to medium-grained sandstone samples show systematic changes in the clast composition. The number of samples is not sufficient to resolve questions about variations in composition as a function of small-scale facies variations, and the samples do not address lateral variations in composition; consequently we examine only major trends in the mean compositions.

The sandstones change upward from lithic arenites to feldspathic arenites. The sandstones

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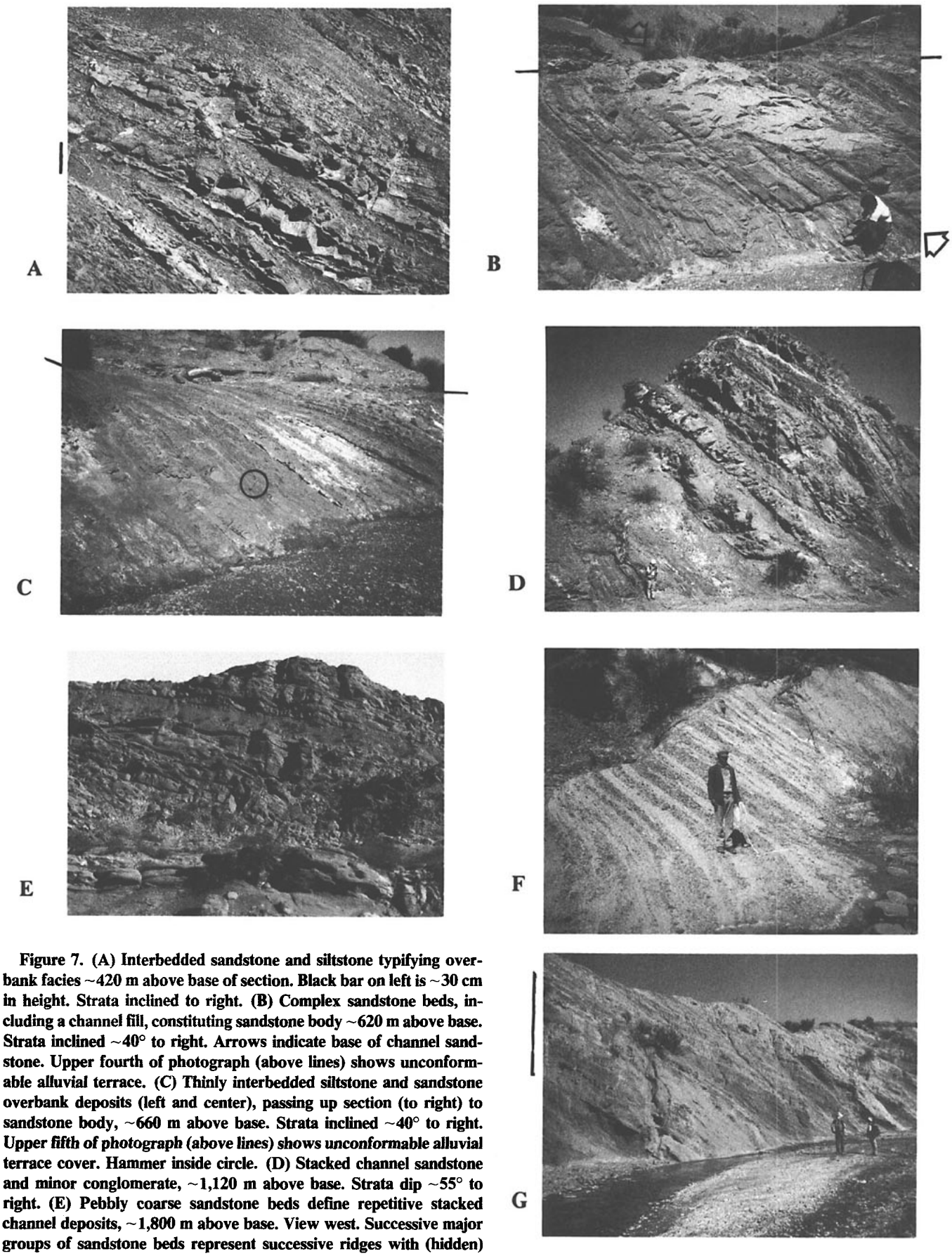


Figure 7. (A) Interbedded sandstone and siltstone typifying overbank facies ~420 m above base of section. Black bar on left is ~30 cm in height. Strata inclined to right. (B) Complex sandstone beds, including a channel fill, constituting sandstone body ~620 m above base. Strata inclined ~40° to right. Arrows indicate base of channel sandstone. Upper fourth of photograph (above lines) shows unconformable alluvial terrace. (C) Thinly interbedded siltstone and sandstone overbank deposits (left and center), passing up section (to right) to sandstone body, ~660 m above base. Strata inclined ~40° to right. Upper fifth of photograph (above lines) shows unconformable alluvial terrace cover. Hammer inside circle. (D) Stacked channel sandstone and minor conglomerate, ~1,120 m above base. Strata dip ~55° to right. (E) Pebbly coarse sandstone beds define repetitive stacked channel deposits, ~1,800 m above base. View west. Successive major groups of sandstone beds represent successive ridges with (hidden) swales between. (F) Sheets and lenses of gravel in a channel deposit, ~4,320 m above base of section. Strata inclined ~40° to right. (G) Conglomeratic channel sandstone (indicated by bar) interbedded with thick poorly sorted sandstone of overbank facies, typical near 4,200 m. Strata inclined ~40° to right.

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of the La Flecha and Las Juntas members have 40%–63% quartz and feldspar plus 31%–46% lithic fragments. Above those levels, the sandstones have high percentages of quartz and feldspar (63%–77%) and fewer total lithic grains (18%–33%) (Fig. 8A). The compositions of the lithic grains change systematically upward through the section (Fig. 8B). At the base, volcanic lithics are abundant (25%–45%), dropping steadily through the La Flecha and Las Juntas members to ~11%, and in higher strata, they fluctuate between 10% and 25% (Fig. 8B). A compensatory variation is seen in sedimentary lithic grains, which are quite low at the base (0.5%), increase steadily through the La Flecha member, and then increase abruptly and remain relatively abundant in higher levels in the section (Fig. 8B).

Paleocurrent indicators have not been systematically studied; however, preliminary observations indicate east-directed transport of sediment in both the Vinchina and Zapallar Formations. This is consistent with the compositions of the lithic clasts, indicating that material was derived from the Frontal Cordillera and Precordillera.

Interpretation of Strata

All of the strata are fluvial deposits. The sandstone bodies of the lower part of the Vinchina Formation, and conglomerate and sandstone bodies of the upper Vinchina and Zapallar Formations, are channel deposits; finer-grained intervals represent overbank facies. The siltstones and comparatively fine-grained sandstones at the base of the section are replaced upward by progressively coarser-textured deposits, implying a long-term transition from low- to high-energy fluvial environments.

The siltstones and paleosols of the basal Vinchina Formation (La Flecha member) represent episodic deposition in overbank areas on an alluvial plain, for which the channels had relatively cohesive banks and comparatively fixed positions. The Las Juntas member corresponds to a set of streams with a more mobile habit, on a low-relief braid plain. The Cayó member marks a time of gradual change to more fixed channel positions, indicated by stacked channel deposits and extended periods of soil formation in overbank areas. The Cayó environment is transitional into the La Brea member of the Vinchina Formation and progressively into the Zapallar Formation, which represent a continuum of depositional environments characteristic of large gravel-bed streams, which may have been perennial but had quite variable discharge. We interpret the Cayó and overlying strata to be streamflow deposits of an alluvial fan. The upward coarsening of both the channel and overbank deposits and the gradual changes in facies characteristics through the Cayó and younger units imply that an alluvial fan prograded more or less steadily across the Las Juntas site.

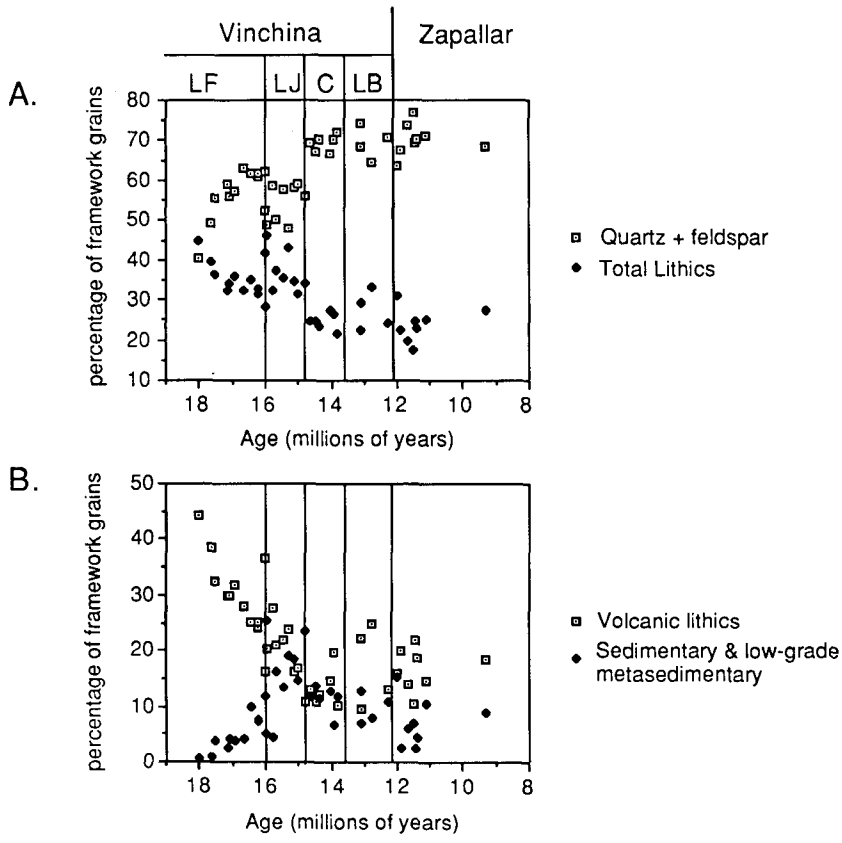


Figure 8. Compositions of sandstone framework grains plotted against age of the stratigraphic horizon (age determined from FIg. 10). (A) Bulk composition of sandstones illustrated by combined monocrystalline quartz and feldspar and by total lithic clasts. (B) Principal lithic clast types, comparing volcanic (derived from the Frontal Cordillera) and sedimentary lithic clasts (may have been derived from both the Precordillera and the Frontal Cordillera). Formations and informal members indicated at top of graph A (LF, La Flecha member; LJ, Las Juntas member; C, Cayó member; LB, La Brea member).

PALEOMAGNETIC RESULTS

Of the 168 sites sampled in the Las Juntas section, 142 sites (85%) yielded interpretable results, 4 sites (2%) were located in strata that were repeated owing to faulting (Figs. 3 and 4), and 24 sites (13%) yielded uninterpretable results. Of the sites used in our interpretation, 124 (87%) are Class I sites and 18 (13%) are Class II sites. A plot of virtual geomagnetic pole (VGP) latitude versus stratigraphic position in the section is abstracted into a standard black-and-white local paleomagnetic column (normal polarity zones in black) (Fig. 5). A total of 31 reversals was encountered in the section. Of the 32 polarity zones thus defined, 29 are represented by more than one sampling site. A graphical reversal test (Fig. 9) shows that the mean orientations of the normally and reversely polarized sample sets are approximately antipodal and supports our contention that the components of magnetization isolated in demagnetization represent a detrital remanent magnetization. Uncertainties in the stratigraphic position of the polarity boundaries are determined by the distances between sample sites that define the reversals; the uncertainty averages ±41 m, with a greater uncertainty in the upper part of the section where fewer lithologies suitable for sampling are present (±24.5 m below 2,000 m in section; ±59 m above 2,000 m).

We correlate the Las Juntas paleomagnetic column to the 18–9 Ma time period (Fig. 10) based on the pattern of reversals, a fission-track date on an interbedded tuff, and a statistical test of the probable duration of the section. Our

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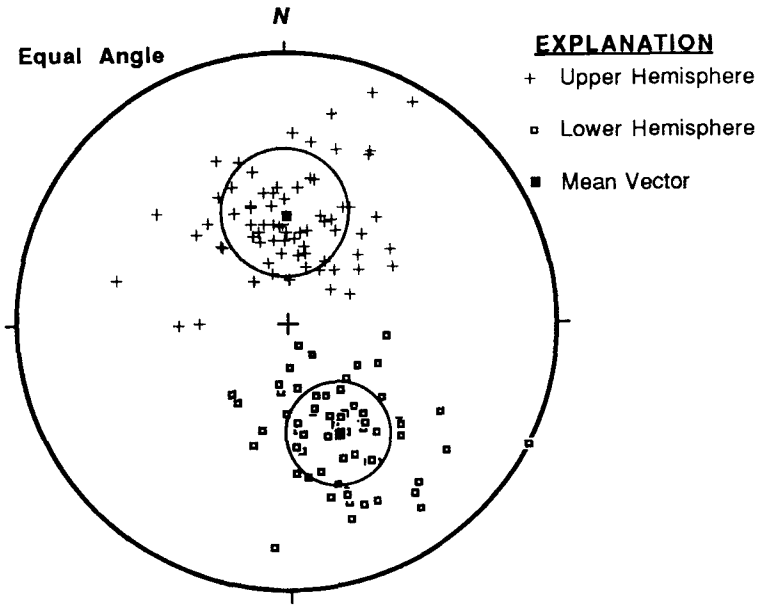


Figure 9. Equal-angle plot of the mean declinations and inclinations of all Neogene Class I sites (Fig. 5) determined using the method of Fisher (1953). The mean declination of the normal set is 001° with a mean inclination of -47° and mean angular deviation of 27°. For the reversely polarized set, the mean declination is 155° and the mean inclination is +42° with a mean angular deviation of 22°. The circles of confidence about the mean orientations (using the mean angular deviation as the radius) of the normally and reversely polarized vectors are approximately antipodal, suggesting that the primary component of magnetization in these strata is a DRM.

correlation with the time scale of Berggren and others (1985) places the base of the section within sea-floor Anomaly 5D, whereas the top is located within Anomaly 5. The reversed polarity zone in which the base lies occurred between 18.1 and 17.9 Ma. The strata at the base of the section are the oldest Neogene continental strata to be reported in the flat-slab region. The best estimate of the age of the top of the section, within a long normal polarity interval, is ~9.3 Ma. One short normal polarity zone, located between Anomalies 5A and 5AA, was not accounted for in the sampling program. The tuff, located at approximately the 2,300 m level (Table 1), yields a fission-track age of 12.1 ± 1.4 Ma (error at 2σ) and coincides well with an age of ~13.0 Ma implied by magnetic polarity data.

An independent evaluation of this interpretation (Table 2) is possible using a statistical model of Neogene magnetic polarity reversals for an exponential distribution of samples in time (Johnson and McGee, 1983). The model estimates that the duration of the Las Juntas section is 6.3 ± 2.0 m.y. The upper boundary of the 2σ error is slightly less than our interpretation of an approximate 8.7 m.y. time span (Fig. 10), implying that our sampling program was very efficient at locating polarity chrons.

The gradational contact between the Vinchina and Zapallar Formations is located at approximately 12 to 12.5 Ma (Figs. 5 and 10 and Table 3), whereas at Vinchina, it is placed at 4.3 ± 1.0 Ma. Both formations at Las Juntas are older than the upper 3,200 m of the Vinchina Formation at Sierra de los Colorados, supporting the contention of Ramos and others (1988) that the upper contact of the Vinchina Formation is profoundly diachronous along strike, and

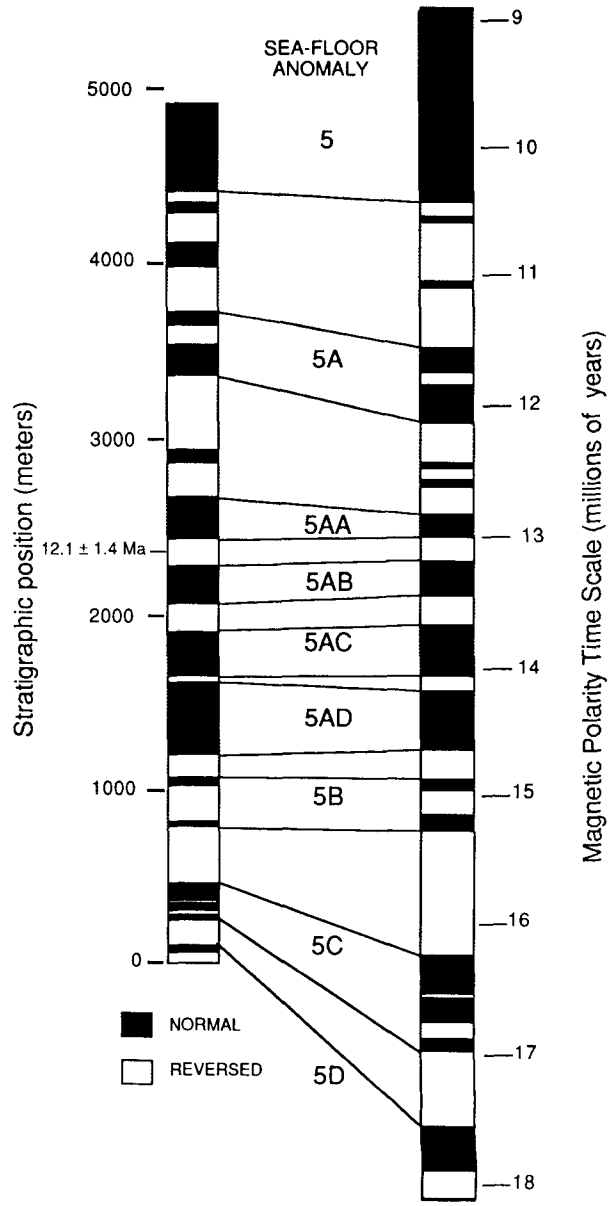


Figure 10. Correlation of the local magnetic polarity stratigraphy (left) of the Las Juntas section (Fig. 5) to the Magnetic Polarity Time Scale (right) of Berggren and others (1985). The local paleomagnetic column is interpreted to represent the 18–9 Ma time frame.

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TABLE 2. ESTIMATE OF THE TIME INTERVAL (Δ) IN LAS JUNTAS SEQUENCE

Number of paleomagnetic sites (N)	Number of reversals found (R)	Probability P = R/(N − 1)	Mean spacing of paleomagnetic sites* S = f(p)	Expected time interval† Δt = SτN (Ma)
142	31	0.2199	0.3676	6.3 ± 2.0 (2σ)

*From Johnson and McGee (1983, equation 7: P = S/(2S + 1). Assuming that samples are distributed with an exponential randomness over the stratigraphic interval.
†An estimate for τ for the late Neogene is 1.2 x 10^5 yr (Johnson and McGee, 1983).

this suggests that the contacts between our informal members may also be diachronous.

Comparison of stratigraphic levels of reversals against absolute age provides a history of the accumulation of strata (Fig. 11); the slope of the resulting curve represents the stratal accumulation rate of the measured section. Because the oldest strata were more deeply buried and more significantly compacted than were younger strata, the accumulation history (Fig. 11, open squares) will be somewhat distorted; however, if the stratal thicknesses are corrected for progressive compaction during burial of the strata (Fig. 11, black diamonds), the estimate of the rate of accumulation is improved (App. A). The average "decompacted" rate of accumulation varied from a minimum of 0.3 mm/a at the base of the section to a maximum in excess of 1.0 mm/a in the middle, and then diminished to 0.4 mm/a at the top.

We examine rate variations in more detail by assessing the slope of the accumulation curve against age (Fig. 12) (note that the rate curve is subjective because the degree of smoothing between original data points is subjective; here third-order polynomials were fit to three separate parts of the curve shown in Fig. 11). The initial increase in rate is gradual between approximately 17 and 16 Ma. By about 15.3 Ma, the rate triples and then varies about a high mean with an extreme of ~1.6 mm/a at ~13 Ma. The rate declines thereafter, with a final slight increase during the youngest interval studied (Fig. 12).

Short-duration variations in the rates (deviations from a smooth line that could be drawn in Fig. 11) may reflect instability in the fluvial system (Johnson and others, 1988), error in positions of polarity boundaries, and/or problems with calculating "decompacted" thicknesses. Little of the variability is due to error in the stratigraphic location of the polarity boundaries. The height of the symbols in Figure 11 (open squares) would mask as much as ±50 m of uncertainty in stratigraphic position, and only one polarity zone boundary (point A) has greater uncertainty (±66 m). Variability introduced in the calculation of original thicknesses is illustrated by the difference in rate implied by point B on the measured-thickness curve (open squares) versus "decompacted" curve (black diamonds). This occurred because the ratio of channel to overbank deposits of the interval of strata above point B is smaller than is typical of adjacent intervals and because overbank and channel deposits compact much differently at shallow burial depths (App. A).

TABLE 3. AGES OF STRATIGRAPHIC CONTACTS IN LAS JUNTAS SECTION

Contact	Stratigraphic level (m)	Age (Ma)
Zapallar/top	4,780	~9.3
La Brea/Zapallar	3,200	12.2
Cayó/La Brea	1,900	13.7
Las Juntas/Cayó	1,100	14.8
La Flecha/Las Juntas	540	16.0
Base/La Flecha	0	~18.0

INTERPRETATION

Our interpretations of the sequence of tectonic events that influenced deposition of these strata are based on paleomagnetic chronology, sedimentary facies analysis, and sand and gravel provenance. The principal features of the sequence are the coarsening-upward texture, the upward decline in volcanic clasts and increase in sedimentary clasts, and the sigmoidal plot of the stratal accumulation history. These changes reflect the initiation of deformation in the Andes west of the Las Juntas site and progressive eastward migration of thrust activity.

We interpret the vertical changes in strata at Las Juntas, described above, in terms of structural history of adjacent regions rather than in terms of autocyclic processes inherent in nonmarine deposition. This interpretation is justified given the temporal and spatial scales involved; the progradation of facies involves thousands of meters of strata and nearly 10 m.y. This exceeds the scale of progradational sequences in tectonically active settings described by Steele and others (1977) and DeCelles and others (1987). Undoubtedly there are autocyclic variations, such as channel migrations and fan incision, superimposed on the large-scale features and demonstrated by second-order variations of the data sets.

The lowest interval of strata (La Flecha member, 18 to 16 Ma) formed while the average rate of accumulation (0.3 mm/a) was slow and reflects a river system with fixed channels and long exposure of overbank areas. The abundance of volcanic lithic grains and paucity of sedimentary lithic grains strongly suggest derivation of the sediment largely from the Frontal Cordillera (Damanti, 1989).

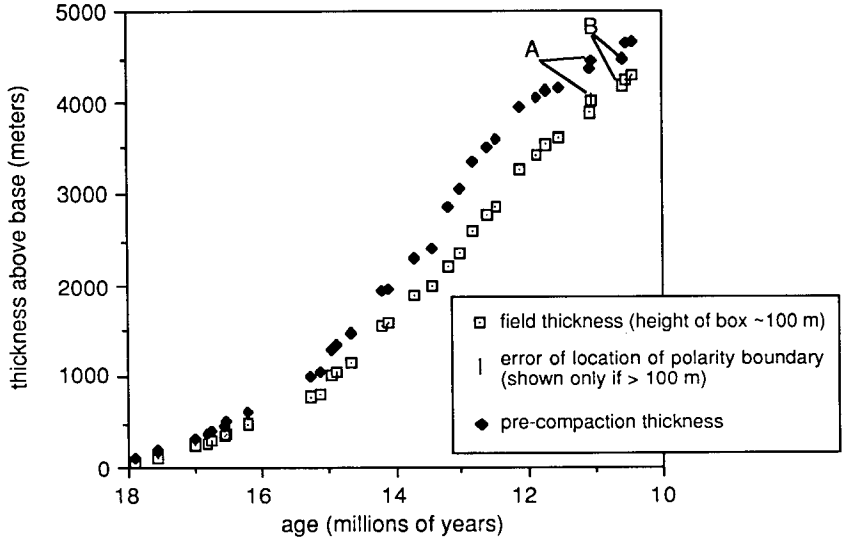


Figure 11. Sediment accumulation history illustrated in a plot of the stratigraphic level of the magnetic reversals (vertical axis) versus the absolute age of the reversals, based on the correlation shown in Figure 10. Curve showing stratal thicknesses corrected for progressive accumulation (black diamonds) converges with field-thickness curve at the stratigraphic level corresponding to top of the original section (not shown, because its exact age is not known).

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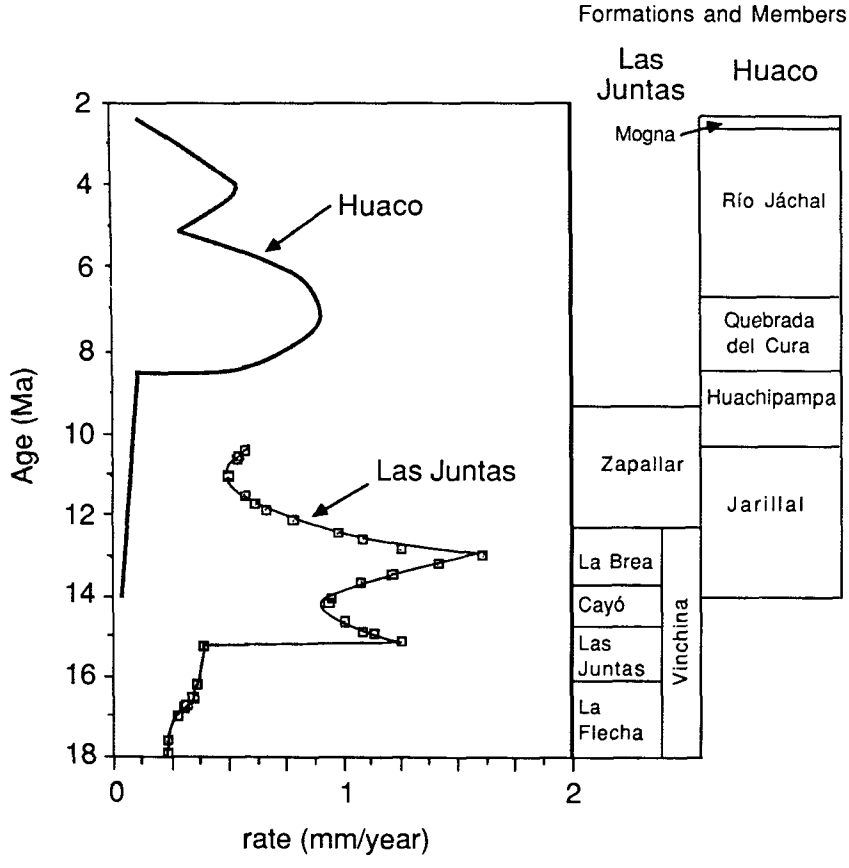


Figure 12. Accumulation rate of sections exposed at Las Juntas and Huaco versus absolute age, using accumulation histories corrected for progressive stratal compaction. The Huaco curve is based on Jordan and others (1988). Huaco location shown in Figure 2.

The Las Juntas member (16 to 14.8 Ma) represents a time of significant change in the drainage system and, by inference, in the structural setting. In combination, the dramatic increases in the rate of accumulation of strata and in the content of sedimentary lithic grains, and increased mobility of the channels, imply that the source area had moved eastward into the Precordillera, slightly increasing the depositional gradient.

After 14.8 Ma, a system of braided streams on a prograding alluvial fan controlled deposition (Cayó member). The low but variable content of volcanic lithic grains implies that the headwaters of the rivers feeding the alluvial fan were largely restricted to the Precordillera (Fig. 2), with possible minor contribution from the Frontal Cordillera or, more likely, with cannibalization of the Miocene strata (for example, Steidtmann and Schmitt, 1988). Starting at about 13.7 Ma, fixed channels on a more proximal part of an alluvial fan deposited gravels and coarse sandstones (La Brea member). This coincides approximately with an increase in the rate of accumulation (Fig. 12). We interpret these changes to mean that the fanhead, and by inference the source of both tectonic loads and detrital material, had migrated even closer to the Las Juntas site, probably signifying eastward migration of thrusts into the central part of the Precordillera.

During Zapallar deposition, from 12.2 to ~9.3 Ma, the rate of accumulation declined. The facies indicate fairly long periods of exposure between times of shifts of channel positions in a coarse, proximal part of an alluvial fan. There are at least three possible interpretations of this interval of time: (1) continued thrusting west of Las Juntas, but with a diminishing rate of structural thickening; (2) initiation of deformation at Las Juntas; or (3) comparative structural quiescence. The first interpretation, that thrusting continued west of Las Juntas, is based on a straightforward interpretation that the rate of accumulation mirrors the rate of structural thickening along thrust faults (Jordan and others, 1988). The second interpretation, that Las Juntas began to experience deformation ~12.2 Ma but that the deformation remained minor until after ~9.3 Ma, is based upon the work of Raynolds and Johnson (1985) in the Siwalik Group foreland basin of Pakistan. Accumulation histories for their sections suggest that intervals of diminishing rates of accumulation foreshadowed deformation of the sections and therefore that the age of an inflection point in the accumulation rate curve marked the beginning of foreshadowing of local deformation. Flemings and Jordan (1987), however, showed that a decreasing rate of accumulation is a natural consequence of cessation of thrusting (option 3 above); the subsidence due to new structural movement ceases, but erosion continues to transfer mass from the mountains to the basin, inducing subsidence. Therefore, episodes of thrusting followed by structural quiescence generate accumulation rate curves with inflections like those described by Raynolds and Johnson (1985), but the time of the inflection point is diachronous across the foreland basin and is not related to local deformation. Thus the declining rate of accumulation at Las Juntas might indicate a time of no thrusting in the central part of the Precordillera. It is interesting that the contact between the two formations mapped by Furque (1972a) coincides with the time that the rate of accumulation decreased.

Because the structural geology of the Las Juntas area has not been studied in detail, the relative sequence of movement of north-northeast–trending folds and thrusts (typical of the central Precordillera) and northwest-trending folds (typical of the Sierras Pampeanas) is not known. Both structural domains, however, clearly involve the strata that we have studied. Therefore, although we are uncertain of the time that deformation at Las Juntas began, there was clearly movement younger than ~9.3 Ma.

Comparison of the chronology of stratigraphic units from Las Juntas (Table 3) to those of Ramos and others (1988) from Sierra de los Colorados and of Johnson and others (1986) from Huaco (Fig. 12) indicates that correlation of lithofacies in the western Sierras Pampeanas and Precordillera is not a reliable chronological approach. Similar-appearing units, which may be located grossly along strike of one another but are separated by mountain blocks, differ in age by many millions of years.

SUMMARY AND COMPARISON TO NAZCA PLATE HISTORY

The new chronological data constrain the sequence and timing of deformation of the Chilean and Argentine Andes Mountains in the northern part of the zone of flat subduction (Fig. 13). Comparison of that deformation history to the independently derived history of the form of the Nazca plate suggests that the decrease in

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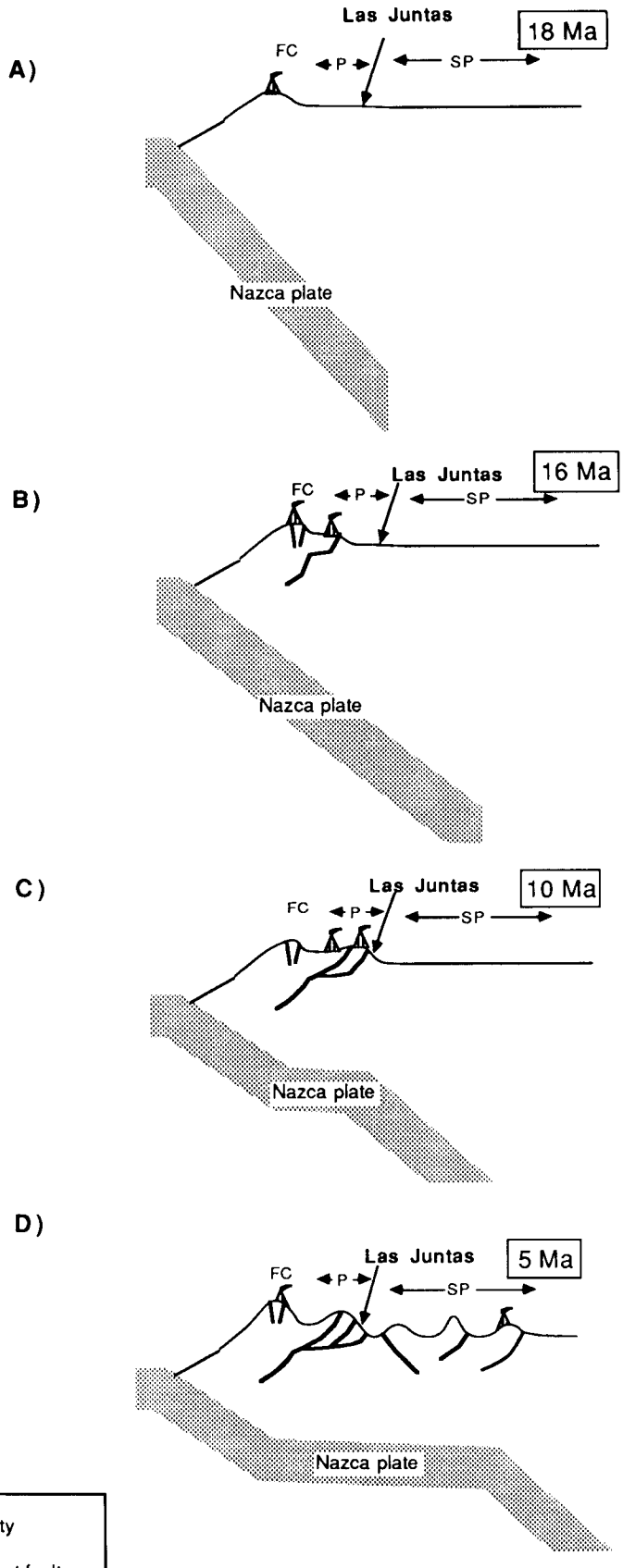


Figure 13. Cross sections of the flat-slab segment of Chile and Argentina and the subducting Nazca plate at successive stages, from oldest (top) to youngest (bottom). The topography and the subducted plate shapes are schematic. Note the position of the Las Juntas site during each stage. In the cartoons, the South American plate is treated as fixed, whereas the Nazca plate is subducted.

angle of subduction of the Nazca plate beneath the South American plate had major consequences for the structural integrity of the continent.

Deformation and uplift in the Frontal Cordillera between 18.9 and 16.6 Ma (Maksaev and others, 1984) were probably responsible for the initiation of foreland basin subsidence and sediment accumulation at Las Juntas about 18 Ma (Fig. 13A). Evidence for uplift in the westernmost Precordillera first appears in the Las Juntas area around 16 Ma (Fig. 13B). Our interpretation suggests that uplift continued to migrate eastward into the central part of the Precordillera until at least 9 Ma when the top of the Las Juntas section was deposited (Fig. 13C). Clearly, deformation migrated farther eastward after 9 Ma, uplifting the Las Juntas section (Fig. 13D). A more complete history of deformation is obtained by combining Las Juntas data with information about other Cenozoic stratigraphic sequences in the area.

Eastward migration of the deformation front is documented along strike of Las Juntas. The chronology of a section exposed about 90 km to the south, at Huaco in the easternmost Precordillera (Fig. 2), was established by Johnson and others (1986), and sedimentary environments and petrology have been studied in detail (Jordan and others, 1988; Beer and Jordan, 1989; Damanti, 1989). The Las Juntas and Huaco sections overlap between 14 and 9 Ma, and accumulation rate curves for these two sections are grossly similar except that they represent different intervals of time (Fig. 12). In the Huaco section, clasts of limestone derived from the central Precordillera first appear in the sedimentary record around 9 Ma and are common by 8.5 Ma (Jordan and others, 1988). Therefore, the limestone-based thrust sheets of the central Precordillera, which are along strike of the Las Juntas area, began to be deformed approximately 9 Ma. Progressive eastward migration of deformation in the Precordillera at the latitude of Huaco is interpreted on the basis of the facies history at Huaco (Jordan and others, 1988; Beer and Jordan, 1989; Damanti, 1989). Deformation of the

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easternmost Precordillera occurred after 2.3 Ma (Johnson and others, 1986; Johnsson and others, 1984).

Assuming that deformation of the Precordillera was roughly contemporaneous along the ~100-km strike distance between Las Juntas and Huaco, we can summarize the history of deformation (Fig. 13). Deformation of the Frontal Cordillera in Chile between 18.9 and 16.6 Ma resulted in initiation of deposition in the Andean foreland. Thin-skinned thrusting of the Precordillera began approximately 16 Ma and has continued to the present. To the degree of resolution permitted by our sampling distribution and the mixture of crosscutting relations and indirect interpretations of deformation, it appears that the eastern front of thrusting migrated progressively, if not steadily, eastward.

The history of deformation east of Las Juntas and Huaco can be derived only from sites located farther east. Miocene strata in the Sierras Pampeanas indicate that deformation within the Sierras Pampeanas is younger than about 8 Ma (Reynolds, 1987; Reynolds and others, 1987b, 1988; Tabbutt and others, 1987b, 1990; Ramos and others, 1988) and that deformation in the Sierras Pampeanas did not progress in a systematic spatial sequence (Jordan and Allmendinger, 1986; Reynolds and others, 1987b). Thus, crustal deformation is not localized along a narrow belt that migrated monotonically toward the foreland; during the past 8 m.y., basement deformation to the east of the Precordillera has been coeval with the thin-skinned deformation of the Precordillera. The existence of metamorphic gravels in strata presumed to be of middle Miocene age at the northern end of the Las Juntas outcrop belt suggests that local basement deformation, typical of the Sierras Pampeanas, began considerably earlier where the main part of the Precordillera thrust system impinged on old basement blocks, as it must have at Sierra Umango (Fig. 2).

In the flat-slab segment, volcanic activity and compressive deformation were complexly related (Fig. 13). A pre–18 Ma volcanic arc, which was typical of relatively high-angle subduction of the Nazca plate beneath the Andes (Kay and others, 1987, 1988), was followed by reverse faulting. Later, thin-skinned thrusting progressed eastward in the Precordillera, contemporaneous with declining volcanic activity in the Frontal Cordillera and with widespread local volcanism within the Precordillera proper (Kay and others, 1988; Ramos and others, 1987, 1990; Tabbutt and others, 1987a, 1990). Thick-skinned shortening and volcanism were also spatially and temporally linked in the Sierras Pampeanas (Kay and others, 1988; Tabbutt and others, 1990).

The volcanic and geochemical data indicate that the subducted plate began to shallow around 16 to 17 Ma that it had a flat part by about 10 Ma, and that the eastern edge of the flat part moved eastward until at least ~5 Ma (Fig. 13) (Jordan and others, 1983b; Kay and others, 1987; Isacks, 1988; Jordan and Gardeweg, 1989). The deformation history developed here reveals that the foreland shortening correlates closely to the geometric history of the subducted plate. The foreland basin began to accumulate strata at Las Juntas soon after high-angle reverse faults began to uplift the Frontal Cordillera, which accompanied or immediately preceded the beginning of plate flattening. Significant thin-skinned shortening accompanied the time of principal transition to a flat geometry. Shortening has continued during the time that flat subduction continued. The locus of supracrustal shortening progressively migrated toward the foreland, but basement deformation has been more broadly distributed across the region of flat subduction.

The coincidence of flattening of the subducted plate and shortening of the foreland is dramatic, suggesting that either the change in subduction geometry, or the flat geometry itself, caused much of the shortening. Deformation in the Andean foreland since 18 Ma has not been restricted to the region of flat subduction, however. In the segment to the north, in which the Nazca plate dips about 30° beneath the Andes, horizontal compression produced in excess of 100 km of shortening during the Neogene, most of it since 10 to 12 Ma (Allmendinger and others, 1983; Jordan and Alonso, 1987). Thus the Miocene and younger shortening documented in the flat-slab region is not uniquely caused by the low-angle subduction but may be partly related to broader-scale facets of the plate margin (Isacks, 1988; Jordan and Gardeweg, 1989). Until more comprehensive and precise dating of the deformation in the northern segment takes place, it will be difficult to compare and contrast the deformation histories in the two segments and to reach conclusions about the relative importance of competing hypotheses bearing on the deformation.

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APPENDIX A. METHOD APPLIED TO "DECOMPACT" THE LAS JUNTAS STRATA

Correction of rock thicknesses to account for progressive compaction of strata ("decompaction") was based on the approach described by Sclater and Christie (1980), using functional relations specific to the sandstones and mudstones at Las Juntas. As part of determination of modal compositions of 38 sandstone samples (Fig. 8), using a petrographic microscope, J. Damanti determined the percentages of framework grains and intergrain volume (cement plus porosity, but in this case porosity is near zero). A plot of intergrain volumes versus stratigraphic position resulted in two functional relationships between depth and degree of compaction of sandstones (which we extrapolate to the conglomerates with sandstone matrix). Above 2,000 m in the measured section (Fig. 4),

$$f = 38.2 \times 10^{(-2.6 \times 10^{-4} m^{-1} \times Z)},$$

and below 2,000 m,

$$f = 41.0 - (6.9 \times 10^{-3}\ m^{-1})Z$$

where f is porosity in percent and Z is depth in meters beneath the top of the section.

For mudstones and poorly sorted overbank sandstones, we measured the densities of 10 rock samples and converted those to porosity (assuming grain density of 2.65 gm/cm^3 for the minerals), resulting in the depth-porosity relationship

$$f = (Z^{-5.92})/(1.63 \times 10^{-3}).$$

These expressions were used to progressively track the decompaction of each magnetic polarity zone using an Excel spreadsheet and a Macintosh computer. We assumed that prior to deformation and erosion, there was an additional 200 m of strata overlying the top of our measured section.

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